

LUCA Health OS — Database Schema Documentation v3

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Database engine: PostgreSQL

Total modules: 14 | **Total tables:** 31 | **Total enums:** 18

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1. Patient Identity

Module purpose: Global patient identity across the entire LUCA ecosystem. A patient logs in once via WhatsApp/phone and can link to multiple doctors' records.

PatientAccount

Stores the master identity for every patient in the system. One `PatientAccount` can be linked to multiple `Patient` records (one per doctor).

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>phone</code>	<code>VARCHAR</code>	NOT NULL	—	Global identifier — WhatsApp number. Unique . Used for OTP login
3	<code>email</code>	<code>VARCHAR</code>	NULL	—	Optional email for notifications. Unique when set
4	<code>passwordHash</code>	<code>VARCHAR</code>	NOT NULL	—	bcrypt/argon2 hash. Used for web portal login
5	<code>fullName</code>	<code>VARCHAR</code>	NOT NULL	—	Patient's full legal name
6	<code>avatarUrl</code>	<code>VARCHAR</code>	NULL	—	Profile picture URL (S3/Cloud Storage)
7	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Account creation timestamp

Business rules:

- A patient registers with their phone number (WhatsApp) — this is the universal identifier
- Email is optional but must be unique when provided
- The phone number is the global key: a patient can have multiple doctors but only one account

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (phone)`
- `UNIQUE (email)` — partial index, nulls not distinct

Zod schema: *(not exposed to frontend directly — managed via auth flow)*

2. Users & Roles

Module purpose: All professional users: doctors, pharmacy/lab providers, and system administrators. Controls authentication, plan tier, and professional branding.

User

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>email</code>	<code>VARCHAR</code>	NOT NULL	—	Login email. Unique
3	<code>passwordHash</code>	<code>VARCHAR</code>	NOT NULL	—	bcrypt/argon2 hash
4	<code>fullName</code>	<code>VARCHAR</code>	NOT NULL	—	Full legal name
5	<code>phone</code>	<code>VARCHAR</code>	NULL	—	Administrative WhatsApp number
6	<code>role</code>	<code>UserRole</code>	NOT NULL	<code>DOCTOR</code>	DOCTOR -> PROVIDER -> ADMIN
7	<code>isActive</code>	<code>BOOLEAN</code>	NOT NULL	<code>true</code>	Soft-delete / account suspension
8	<code>planType</code>	<code>PlanType</code>	NOT NULL	<code>FREE</code>	FREE -> PRO -> ENTERPRISE
9	<code>specialty</code>	<code>VARCHAR</code>	NULL	—	e.g. "Cardiología", "Pediatria" (DOCTOR only)
10	<code>logoUrl</code>	<code>VARCHAR</code>	NULL	—	Professional logo for prescriptions/certificates
11	<code>signatureUrl</code>	<code>VARCHAR</code>	NULL	—	Digital signature image URL
12	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Registration date
13	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last profile update

Business rules:

- `role=DOCTOR` -> medical professional, can manage patients, write prescriptions, conduct consultations
- `role=PROVIDER` -> pharmacy or laboratory, must have a linked `ProviderProfile`
- `role=ADMIN` -> LUCA platform administrator
- `planType` controls feature access: FREE (basic), PRO (full features), ENTERPRISE (multi-clinic, custom branding)

- `specialty`, `logoUrl`, `signatureUrl` are only relevant for DOCTOR role

Indexes:

- PRIMARY KEY (`id`)
- UNIQUE (`email`)

Enum: UserRole

VALUE	DESCRIPTION
DOCTOR	Medical professional – can manage patients, prescribe, consult
PROVIDER	Pharmacy or laboratory – marketplace participant
ADMIN	LUCA platform administrator

Enum: PlanType

VALUE	DESCRIPTION
FREE	Basic tier – limited patients, no marketplace
PRO	Full tier – unlimited patients, marketplace, templates
ENTERPRISE	Multi-clinic – custom branding, clinic management, API access

3. Patient Records

Module purpose: Medical CRM. Each doctor manages their own patient records. A patient can exist in multiple doctors' `Patient` tables linked to the same `PatientAccount`.

Patient

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Doctor who owns this record
3	<code>patientAccountId</code>	<code>VARCHAR</code>	NULL	—	FK -> PatientAccount. Global patient link
4	<code>firstName</code>	<code>VARCHAR</code>	NOT NULL	—	Patient's first name
5	<code>lastName</code>	<code>VARCHAR</code>	NOT NULL	—	Patient's last name
6	<code>nationalId</code>	<code>VARCHAR</code>	NOT NULL	—	National ID / DNI / passport
7	<code>birthDate</code>	<code>DATETIME</code>	NOT NULL	—	Date of birth
8	<code>gender</code>	<code>Gender</code>	NOT NULL	—	MALE -> FEMALE -> OTHER
9	<code>email</code>	<code>VARCHAR</code>	NULL	—	Contact email
10	<code>phone</code>	<code>VARCHAR</code>	NULL	—	Contact phone
11	<code>address</code>	<code>VARCHAR</code>	NULL	—	Physical address
12	<code>city</code>	<code>VARCHAR</code>	NULL	—	City of residence
13	<code>emergencyContactName</code>	<code>VARCHAR</code>	NULL	—	v3 🟡 Emergency contact full name
14	<code>emergencyContactPhone</code>	<code>VARCHAR</code>	NULL	—	v3 🟡 Emergency contact phone number
15	<code>accessCode</code>	<code>VARCHAR</code>	NULL	—	Unique code for patient portal access. Unique
16	<code>lastLogin</code>	<code>DATETIME</code>	NULL	—	Last patient portal login timestamp
17	<code>bloodType</code>	<code>VARCHAR</code>	NULL	—	A+/A-/B+/B-/AB+/AB-/O+/O-
18	<code>allergies</code>	<code>VARCHAR</code>	NULL	—	Known allergies (comma-separated or JSON array)

19	<code>chronicConditions</code>	<code>VARCHAR</code>	<code>NULL</code>	—	Chronic conditions summary
20	<code>privateNotes</code>	<code>TEXT</code>	<code>NULL</code>	—	Doctor-only private notes (not visible to patient)
21	<code>createdAt</code>	<code>TIMESTAM P</code>	<code>NOT NULL</code>	<code>NOW (</code> <code>)</code>	Record creation
22	<code>updatedAt</code>	<code>TIMESTAM P</code>	<code>NOT NULL</code>	<code>NOW (</code> <code>)</code>	Last update

Business rules:

- `userId` defines ownership: only the creating doctor can see this record
- `patientAccountId` links to the global account — enables cross-doctor data sharing with patient consent
- `accessCode` is auto-generated, used for patient portal login without password
- `privateNotes` are NEVER visible to the patient via the portal
- `emergencyContact*` fields were added in v3 — already present in frontend Zod schema

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId)`
- `INDEX (patientAccountId)`
- `UNIQUE (accessCode)`

Enum: Gender

VALUE	DESCRIPTION
<code>MAL E</code>	Male
<code>FEMA LE</code>	Female
<code>OTHE R</code>	Other / non-binary

Zod schema: `src/features/patients/schemas.ts`

4. Clinical Module

Module purpose: Core clinical workflow: dynamic form templates -> consultations with SOAP notes -> vital signs -> lab requests -> appointments.

FormTemplate

Dynamic form builder — doctors can create custom clinical history templates or use global ones by specialty.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NULL	—	FK -> User. NULL = global template, NOT NULL = personal
3	<code>specialty</code>	<code>VARCHAR</code>	NULL	—	e.g. "Cardiología". For global templates
4	<code>schemaJson</code>	<code>JSON</code>	NOT NULL	—	Array of input definitions: <code>[{name, label, type, options?}]</code>
5	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Creation timestamp

Business rules:

- `userId=NULL` -> global template curated by LUCA for a specialty
- `userId=NOT NULL` -> personal template created by a doctor (drag & drop)
- `schemaJson` structure: `[{name: "bpm", label: "Frecuencia cardíaca", type: "number", unit: "bpm"}]`
- Supported field types: text, number, select, multiselect, date, textarea, image

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId)`
- `INDEX (specialty) WHERE userId IS NULL`

Consultation

The core clinical encounter. Every doctor-patient interaction is a Consultation. Supports dynamic form data via `dynamicData`.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Doctor conducting the consultation

3	<code>patientId</code>	<code>VARCHAR</code> <code>R</code>	NOT NULL	—	FK -> Patient. Patient being seen
4	<code>clinicId</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	FK -> Clinic. NULL if private practice
5	<code>formTemplateId</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	FK -> FormTemplate. Template used for this consult
6	<code>date</code>	<code>DATETIME</code>	NOT NULL	—	Consultation date and time
7	<code>status</code>	<code>VARCHAR</code> <code>R</code>	NOT NULL	<code>pending</code>	v2: pending -> in-progress -> completed -> cancelled
8	<code>reason</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	S (Subjective): Chief complaint / reason for visit
9	<code>physicalExam</code>	<code>TEXT</code>	NULL	—	O (Objective): Physical examination findings
10	<code>diagnosis</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	A (Assessment): Diagnosis
11	<code>treatmentPlan</code>	<code>TEXT</code>	NULL	—	P (Plan): Treatment plan
12	<code>dynamicData</code>	<code>JSON</code>	NULL	—	Custom form responses: <code>{bpm: 85, perimetro: 35}</code>

Business rules:

- SOAP note structure: Subjective (reason), Objective (physicalExam + vital signs), Assessment (diagnosis), Plan (treatmentPlan)
- `dynamicData` stores form template responses — keyed by form field `name`
- Status workflow: `pending` (scheduled) -> `in-progress` (doctor started) -> `completed` (finished) -> `cancelled` (not attended)
- A consultation can have exactly one `VitalSign` and one `LabRequest`

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId, date)`
- `INDEX (patientId)`
- `INDEX (clinicId)`
- `INDEX (status)`

Appointment

Scheduling/agenda system. Doctors manage their appointment calendar.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)

2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
3	<code>doctorId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User (must have role=DOCTOR)
4	<code>clinicId</code>	<code>VARCHAR</code>	NULL	—	FK -> Clinic. NULL if private practice
5	<code>date</code>	<code>DATE</code>	NOT NULL	—	Appointment date
6	<code>time</code>	<code>VARCHAR</code>	NOT NULL	—	Time slot, e.g. "09:30"
7	<code>type</code>	<code>VARCHAR</code>	NOT NULL	—	e.g. "Control general", "Cardiología"
8	<code>status</code>	<code>VARCHAR</code>	NOT NULL	<code>pending</code>	pending -> in-progress -> completed -> cancelled
9	<code>notes</code>	<code>TEXT</code>	NULL	—	Internal doctor notes
10	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Creation
11	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last update

Business rules:

- Doctor (and clinic) calendar: appointments are scoped to doctor+date
- Status drives the dashboard queue: `pending` appears in "upcoming", `in-progress` in "now"
- No double-booking enforced at application level

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (doctorId, date)`
- `INDEX (patientId)`
- `INDEX (clinicId)`

VitalSign

Patient vitals taken during a consultation. 1:1 relationship with Consultation.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
3	<code>consultationId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Consultation. Unique (1:1)
4	<code>weight</code>	<code>FLOAT</code>	NULL	—	kg
5	<code>height</code>	<code>FLOAT</code>	NULL	—	cm
6	<code>systolicBP</code>	<code>INT</code>	NULL	—	Systolic blood pressure (mmHg)

7	<code>diastolicBP</code>	<code>INT</code>	<code>NULL</code>	—	Diastolic blood pressure (mmHg)
8	<code>heartRate</code>	<code>INT</code>	<code>NULL</code>	—	bpm
9	<code>temperature</code>	<code>FLOAT</code>	<code>NULL</code>	—	°C
10	<code>oxygenSat</code>	<code>INT</code>	<code>NULL</code>	—	SpO ₂ percentage
11	<code>date</code>	<code>DATETIME</code>	<code>NOT NULL</code>	—	Measurement date/time

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (consultationId)`
- `INDEX (patientId)`

LabRequest

Laboratory test order linked to a consultation. 1:1 relationship with Consultation.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code> <code>R</code>	<code>NOT NULL</code>	—	Primary key (UUID v7)
2	<code>consultationId</code>	<code>VARCHAR</code> <code>R</code>	<code>NOT NULL</code>	—	FK -> Consultation. Unique (1:1)
3	<code>examsList</code>	<code>VARCHAR</code> <code>R</code>	<code>NOT NULL</code>	—	JSON array of exam names: <code>["Hemograma", "Glicemia"]</code>
4	<code>instructions</code>	<code>VARCHAR</code> <code>R</code>	<code>NULL</code>	—	Special instructions for patient/lab
5	<code>isCompleted</code>	<code>BOOLEAN</code>	<code>NOT NULL</code>	<code>false</code>	All results received?

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (consultationId)`

5. Prescriptions

Module purpose: Complete prescription workflow: medication catalog -> prescription templates -> live prescriptions with items.

Medication

Global and personal medication catalog. Global medications are curated by LUCA; doctors can add personal entries.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NULL	—	FK -> User. NULL = global catalog, NOT NULL = doctor's personal
3	<code>activePrinciple</code>	<code>VARCHAR</code>	NOT NULL	—	Active ingredient, e.g. "Amoxicilina"
4	<code>concentration</code>	<code>VARCHAR</code>	NOT NULL	—	e.g. "500mg", "4mg/ml", "1g"
5	<code>presentation</code>	<code>PresentationEnum</code>	NOT NULL	—	CAPSULA -> TABLETA -> JARABE -> GOTAS -> AMPOLLA -> CREMA
6	<code>administrationRoute</code>	<code>AdministrationRouteEnum</code>	NOT NULL	—	ORAL -> INTRAVENOSA -> INTRAMUSCULAR -> TOPICA -> OFTALMICA
7	<code>commercialName</code>	<code>VARCHAR</code>	NULL	—	Brand name, e.g. "Amoxil"
8	<code>requiresPrescription</code>	<code>BOOLEAN</code>	NOT NULL	<code>true</code>	v3  Does it require a medical prescription?
9	<code>contraindications</code>	<code>TEXT</code>	NULL	—	v3  Known contraindications
10	<code>isActive</code>	<code>BOOLEAN</code>	NOT NULL	<code>true</code>	Soft-delete / deprecate medication
11	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Creation
12	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last update

Business rules:

- Unique constraint: `(userId, activePrinciple, concentration, presentation, administrationRoute)` — same doctor can't duplicate
- Global catalog: `userId=NULL`, visible to all doctors

- Personal catalog: `userId=NOT NULL` , visible only to that doctor
- Seed data: 12 common medications included in `DATABASE_PATCH_v2.sql`

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (userId, activePrinciple, concentration, presentation, administrationRoute)`
- `INDEX (userId)`

Zod schema: `src/features/medications/schemas.ts`

Enum: PresentationEnum

VALUE	LABEL (ES)	DESCRIPTION
CAPS ULA	Cápsula	Capsule
TABL ETA	Tableta	Tablet
JARA BE	Jarabe	Syrup
GOTA S	Gotas	Drops
AMPO LLA	Ampolla	Ampoule / vial
CREM A	Crema	Cream / ointment

Enum: AdministrationRouteEnum

VALUE	LABEL (ES)	DESCRIPTION
ORA L	Oral	By mouth
INTR AVEN OSA	Intravenosa	Intravenous
INTR AMUS CULA R	Intramuscular	Intramuscular injection
TOPI CA	Tópica	Topical application
OFTA LMIC	Oftálmica	Ophthalmic (eye)

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Prescription

A live prescription issued by a doctor. Can be linked to a consultation (but not required).

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Prescribing doctor
3	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
4	<code>consultationId</code>	<code>VARCHAR</code>	NULL	—	FK -> Consultation. Optional link. Unique
5	<code>clinicId</code>	<code>VARCHAR</code>	NULL	—	FK -> Clinic. NULL if private practice
6	<code>date</code>	<code>DATETIME</code>	NOT NULL	—	Prescription date
7	<code>expirationDate</code>	<code>DATETIME</code>	NOT NULL	—	Expiration date
8	<code>notes</code>	<code>VARCHAR</code>	NULL	—	General notes
9	<code>publicToken</code>	<code>VARCHAR</code>	NOT NULL	—	Shareable token for pharmacy verification. Unique
10	<code>status</code>	<code>RxStatus</code>	NOT NULL	—	ACTIVE -> CANCELLED -> EXPIRED

Business rules:

- `publicToken` is a unique, shareable identifier — patients share this with pharmacies (WhatsApp/link)
- Status transitions: ACTIVE -> CANCELLED (doctor revokes) or ACTIVE -> EXPIRED (past expirationDate)
- If linked to a consultation, it's 1:1 (unique constraint)

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (consultationId)` — partial index, nulls not distinct
- `UNIQUE (publicToken)`
- `INDEX (userId)`
- `INDEX (patientId)`

PrescriptionItem

Individual medication line item within a prescription.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
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1	<code>id</code>	<code>VARCHAR</code> <code>R</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>prescriptionId</code>	<code>VARCHAR</code> <code>R</code>	NOT NULL	—	FK -> Prescription
3	<code>medicationId</code>	<code>VARCHAR</code> <code>R</code>	NOT NULL	—	FK -> Medication. v2: replaced free-text <code>medication</code>
4	<code>dose</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	e.g. "1 cápsula"
5	<code>frequency</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	e.g. "Cada 8 horas"
6	<code>duration</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	e.g. "7 días"
7	<code>quantity</code>	<code>INT</code>	NOT NULL	<code>1</code>	v2: Number of units (boxes, blisters)
8	<code>notes</code>	<code>VARCHAR</code> <code>R</code>	NULL	—	v2: Doctor observations for this specific item

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (prescriptionId)`
- `INDEX (medicationId)`

Enum: RxStatus

VALUE	DESCRIPTION
<code>ACTIVE</code>	Prescription is valid and can be dispensed
<code>CANCELLED</code>	Revoked by the prescribing doctor
<code>EXPIRED</code>	Past the expiration date

PrescriptionTemplate

Saved prescription templates for quick reuse (e.g. "Post-operatorio estándar").

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Doctor who owns the template
3	<code>title</code>	<code>VARCHAR</code>	NOT NULL	—	Template name, e.g. "Post-cirugía abdominal"

Indexes:

- PRIMARY KEY (id)
- INDEX (userId)

TemplateItem

Medication item within a prescription template.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	VARCHAR	NOT NULL	—	Primary key (UUID v7)
2	templateId	VARCHAR	NOT NULL	—	FK -> PrescriptionTemplate
3	medicationId	VARCHAR	NOT NULL	—	FK -> Medication. v2: replaced free-text
4	dose	VARCHAR	NULL	—	e.g. "1 cápsula"
5	frequency	VARCHAR	NULL	—	e.g. "Cada 8 horas"
6	duration	VARCHAR	NULL	—	e.g. "7 días"

Indexes:

- PRIMARY KEY (id)
 - INDEX (templateId)
 - INDEX (medicationId)
-

6. Marketplace

Module purpose: B2B marketplace connecting patients/doctors with pharmacies and laboratories. Quote request -> offer workflow.

ProviderProfile

Business profile for pharmacies and laboratories participating in the marketplace.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Unique (1:1)
3	<code>type</code>	<code>ProviderType</code>	NOT NULL	—	PHARMACY -> LABORATORY
4	<code>commercialName</code>	<code>VARCHAR</code>	NOT NULL	—	Business name
5	<code>rif</code>	<code>VARCHAR</code>	NOT NULL	—	Tax ID (Venezuela RIF)
6	<code>address</code>	<code>VARCHAR</code>	NOT NULL	—	Physical address
7	<code>city</code>	<code>VARCHAR</code>	NOT NULL	—	City
8	<code>state</code>	<code>VARCHAR</code>	NOT NULL	—	State / province
9	<code>phone</code>	<code>VARCHAR</code>	NOT NULL	—	Business phone
10	<code>isOpen</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	Currently accepting orders?
11	<code>isVerified</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	KYC verified?

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (userId)`

Enum: ProviderType

VALUE	DESCRIPTION
<code>PHARMACY</code>	Retail pharmacy / drugstore
<code>LABORATORY</code>	Clinical laboratory

QuoteRequest

A request for medication/lab quotes, typically created from a prescription.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>prescriptionId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Prescription
3	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
4	<code>city</code>	<code>VARCHAR</code>	NOT NULL	—	Target city for delivery/pickup
5	<code>status</code>	<code>QuoteStatus</code>	NOT NULL	—	OPEN -> CLOSED
6	<code>createdAt</code>	<code>DATETIME</code>	NOT NULL	<code>NOW ()</code>	Request creation

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (prescriptionId)`
- `INDEX (patientId)`
- `INDEX (status)`

QuoteOffer

A provider's price offer for a quote request.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>quoteRequestId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> QuoteRequest
3	<code>providerId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> ProviderProfile
4	<code>price</code>	<code>FLOAT</code>	NOT NULL	—	Total price
5	<code>currency</code>	<code>VARCHAR</code>	NOT NULL	<code>"USD"</code>	Currency code
6	<code>availability</code>	<code>VARCHAR</code>	NULL	—	e.g. "Entrega en 24h", "Retiro en sucursal"
7	<code>comments</code>	<code>VARCHAR</code>	NULL	—	Provider notes
8	<code>createdAt</code>	<code>DATETIME</code>	NOT NULL	<code>NOW ()</code>	Offer creation

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (quoteRequestId)`
- `INDEX (providerId)`

Enum: QuoteStatus

VALUE	DESCRIPTION
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Accepting offers from providers

CLOSED

Request fulfilled or expired

7. System

Module purpose: Cross-cutting system features: notifications, medical documents, and patient follow-ups.

Notification

In-app notification system.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Recipient
3	<code>type</code>	<code>NotifType</code>	NOT NULL	—	Notification category
4	<code>title</code>	<code>VARCHAR</code>	NOT NULL	—	Notification title
5	<code>message</code>	<code>VARCHAR</code>	NOT NULL	—	Notification body
6	<code>isRead</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	Read status
7	<code>link</code>	<code>VARCHAR</code>	NULL	—	Deeplink URL
8	<code>createdAt</code>	<code>DATETIME</code>	NOT NULL	<code>NOW()</code>	Creation timestamp

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId, isRead)`
- `INDEX (createdAt)`

Enum: NotifType

VALUE	DESCRIPTION
<code>SYSTEM</code>	General platform notification
<code>NEW_QUOTE_REQUEST</code>	New quote request in marketplace (-> pharmacies)
<code>QUOTE_RECEIVED</code>	Quote received (-> patient/doctor)
<code>FOLLOW_UP</code>	Patient follow-up reminder

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MedicalDocument

Legal/medical documents: certificates, referrals, reports.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	VARCHAR	NOT NULL	—	Primary key (UUID v7)
2	userId	VARCHAR	NOT NULL	—	FK -> User. Issuing doctor
3	patientId	VARCHAR	NOT NULL	—	FK -> Patient
4	type	DocType	NOT NULL	—	CERTIFICATE -> REFERRAL -> REPORT
5	content	TEXT	NOT NULL	—	Document body (HTML or plain text)
6	publicToken	VARCHAR	NOT NULL	—	Shareable verification token. Unique
7	createdAt	DATETIME	NOT NULL	NOW ()	Creation timestamp

Indexes:

- PRIMARY KEY (id)
- UNIQUE (publicToken)
- INDEX (userId)
- INDEX (patientId)

Enum: DocType

VALUE	DESCRIPTION
CERTIFIC	Medical certificate (work/school)
REFERRAL	Inter-doctor referral
REPORT	Medical report

FollowUp

Scheduled patient follow-up tracking.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	id	VARCHAR	NOT NULL	—	Primary key (UUID v7)
2	userId	VARCHAR	NOT NULL	—	FK -> User. Doctor

3	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
4	<code>scheduledDate</code>	<code>DATETIME</code>	NOT NULL	—	When to follow up
5	<code>status</code>	<code>FollowStatus</code>	NOT NULL	—	PENDING -> SENT -> RESPONDED
6	<code>response</code>	<code>VARCHAR</code>	NULL	—	Patient's WhatsApp response

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId, scheduledDate)`
- `INDEX (patientId)`

Enum: FollowStatus

VALUE	DESCRIPTION
<code>PENDING</code>	Follow-up scheduled, not yet sent
<code>SENT</code>	WhatsApp message sent to patient
<code>RESPONDED</code>	Patient responded

8. Medical Background

Module purpose: Deep clinical history — antecedents, surgical history, family history, lifestyle, obstetric history, vaccinations.

MedicalBackground

Key chronic conditions and past hospitalizations. 1:1 with Patient.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient. Unique (1:1)
3	<code>hasDiabetes</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	Diabetes diagnosis
4	<code>hasHypertension</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	Hypertension diagnosis
5	<code>hasAsthma</code>	<code>BOOLEAN</code>	NOT NULL	<code>false</code>	Asthma diagnosis
6	<code>otherConditions</code>	<code>TEXT</code>	NULL	—	Other chronic conditions (free text)
7	<code>pastHospitalizations</code>	<code>TEXT</code>	NULL	—	Past hospitalizations summary

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (patientId)`

SurgicalHistory

Past surgical procedures.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
3	<code>procedure</code>	<code>VARCHAR</code>	NOT NULL	—	Procedure name, e.g. "Apendicectomía"
4	<code>date</code>	<code>DATETIME</code>	NOT NULL	—	Surgery date
5	<code>hospital</code>	<code>VARCHAR</code>	NULL	—	Hospital where performed
6	<code>notes</code>	<code>VARCHAR</code>	NULL	—	Additional notes

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (patientId)`

FamilyHistory

Family medical history — hereditary conditions.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code> R	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code> R	NOT NULL	—	FK -> Patient
3	<code>condition</code>	<code>VARCHAR</code> R	NOT NULL	—	Condition name, e.g. "Diabetes tipo 2"
4	<code>relationship</code>	<code>VARCHAR</code> R	NOT NULL	—	Family relationship, e.g. "Madre", "Abuelo paterno"
5	<code>note</code>	<code>VARCHAR</code> R	NULL	—	Additional details

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (patientId)`

Lifestyle

Patient lifestyle factors. 1:1 with Patient.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code> R	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code> R	NOT NULL	—	FK -> Patient. Unique (1:1)
3	<code>smokingStatus</code>	<code>VARCHAR</code> R	NULL	—	e.g. "No fumador", "Fumador activo", "Ex-fumador"
4	<code>alcoholConsumption</code>	<code>VARCHAR</code> R	NULL	—	e.g. "No consume", "Ocasional", "Frecuente"
5	<code>activityLevel</code>	<code>VARCHAR</code> R	NULL	—	e.g. "Sedentario", "Moderado", "Activo"
6	<code>dietType</code>	<code>VARCHAR</code> R	NULL	—	e.g. "Omnívoro", "Vegetariano", "Vegano"

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (patientId)`

ObstetricHistory

Gynecological/obstetric history. 1:1 with Patient.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient. Unique (1:1)
3	<code>lastPeriodDate</code>	<code>DATETIME</code>	NULL	—	FUM / last menstrual period
4	<code>pregnancies</code>	<code>INT</code>	NULL	—	G (Gravida) — total pregnancies
5	<code>births</code>	<code>INT</code>	NULL	—	P (Para) — births
6	<code>cesareans</code>	<code>INT</code>	NULL	—	C (Cesarean) — C-sections
7	<code>abortions</code>	<code>INT</code>	NULL	—	A (Abortions) — miscarriages/abortions
8	<code>contraceptiveMethod</code>	<code>VARCHAR</code>	NULL	—	Current contraceptive method

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (patientId)`

Vaccination

Immunization records.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
3	<code>vaccine</code>	<code>VARCHAR</code>	NOT NULL	—	Vaccine name, e.g. "COVID-19 Pfizer", "Influenza"
4	<code>doseNumber</code>	<code>INT</code>	NOT NULL	—	Dose number (1, 2, booster...)
5	<code>date</code>	<code>DATETIME</code>	NOT NULL	—	Administration date

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (patientId)`

9. Institutional

Module purpose: Multi-clinic support. Clinics have members (doctors, admins, receptionists) with role-based access.

Clinic

Healthcare institution profile.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>name</code>	<code>VARCHAR</code>	NOT NULL	—	Clinic / hospital name
3	<code>rif</code>	<code>VARCHAR</code>	NOT NULL	—	Tax ID (Venezuela RIF)
4	<code>address</code>	<code>VARCHAR</code>	NOT NULL	—	Physical address
5	<code>logoUrl</code>	<code>VARCHAR</code>	NULL	—	Clinic logo
6	<code>website</code>	<code>VARCHAR</code>	NULL	—	Website URL
7	<code>phone</code>	<code>VARCHAR</code>	NOT NULL	—	Main phone
8	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Creation
9	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last update

Indexes:

- `PRIMARY KEY (id)`

ClinicMember

Membership linking users to clinics with role-based access.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>clinicId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Clinic
3	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User
4	<code>role</code>	<code>ClinicRole</code>	NOT NULL	<code>DOCTOR</code>	OWNER -> ADMIN -> DOCTOR -> RECEPTIONIST
5	<code>isActive</code>	<code>BOOLEAN</code>	NOT NULL	<code>true</code>	Active membership?

Indexes:

- PRIMARY KEY (id)
- INDEX (clinicId)
- INDEX (userId)
- UNIQUE (clinicId, userId)

Enum: ClinicRole

VALUE	PERMISSIONS
OWNER	Full clinic control – billing, member management, analytics
ADMIN	Administrative – manage appointments, members, reports
DOCTOR	Clinical – see own patients, prescribe, consult
RECEPTIONIST	Front desk – schedule appointments, check-in patients

10. Verification (KYC)

Module purpose: Know Your Customer document verification for doctors, pharmacies, and clinics.

VerificationDocument

Uploaded documents for identity/business verification.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Who uploaded
3	<code>type</code>	<code>DocVerificationType</code>	NOT NULL	—	Document category
4	<code>fileUrl</code>	<code>VARCHAR</code>	NOT NULL	—	URL to file (S3/Google Cloud)
5	<code>status</code>	<code>VerificationStatus</code>	NOT NULL	<code>PENDING</code>	PENDING -> APPROVED -> REJECTED
6	<code>comments</code>	<code>TEXT</code>	NULL	—	Rejection reason if applicable
7	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Upload date
8	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last status change

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId, type)`

Enum: DocVerificationType

VALUE	REQUIRED BY
<code>MEDICAL_LICENSE</code>	DOCTOR — medical degree or college certification
<code>NATIONAL_ID</code>	All users — national ID card

BUSINESS_RIF	PROVIDER, CLINIC — tax registration
COMMERCIAL_REGISTRATION	PROVIDER, CLINIC — business registration

Enum: VerificationStatus

VALUE	DESCRIPTION
PENDING	Awaiting review
APPROVED	Document verified
REJECTED	Document rejected (see comments)

11. Pharmacy Inventory v3

Module purpose: Real-time inventory management for pharmacies. Tracks stock levels, batches, expiration dates, and pricing. Enables accurate quote generation in the marketplace.

PharmacyInventory

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>providerId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> ProviderProfile. Pharmacy owning this stock
3	<code>medicationId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Medication. Which medication
4	<code>stock</code>	<code>INT</code>	NOT NULL	0	Current units available
5	<code>minStockAlert</code>	<code>INT</code>	NULL	10	Threshold that triggers low-stock alert
6	<code>batchNumber</code>	<code>VARCHAR</code>	NULL	—	Batch/lot number for traceability
7	<code>expirationDate</code>	<code>DATE</code>	NULL	—	Expiration date of this batch
8	<code>unitPrice</code>	<code>DECIMAL(10, 2)</code>	NULL	—	Selling price per unit (used for auto-quoting)
9	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Record creation
10	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last update

Business rules:

- Unique constraint: `(providerId, medicationId, batchNumber)` — same pharmacy can't have two records for the same batch
- When `stock <= minStockAlert`, the pharmacy receives a "low stock" notification
- `unitPrice` is used to auto-generate quote offers in the marketplace
- `batchNumber` enables recall tracking in case of medication alerts
- Pharmacies update `stock` after each sale/dispensation

Indexes:

- `PRIMARY KEY (id)`
- `UNIQUE (providerId, medicationId, batchNumber)`
- `INDEX (providerId)`
- `INDEX (medicationId)`

- `INDEX (expirationDate)` WHERE expirationDate IS NOT NULL — for expiration alerts

Zod schema: `src/features/inventory/schemas.ts`

12. Lab Results v3

Module purpose: Complements `LabRequest` by storing actual laboratory test results. Supports structured JSON results for flexible test types plus PDF uploads.

LabResult

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>labRequestId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> LabRequest. Unique (1:1)
3	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient. Denormalized for efficient queries
4	<code>fileUrl</code>	<code>VARCHAR</code>	NULL	—	URL to the full PDF report (S3)
5	<code>resultJson</code>	<code>JSONB</code>	NULL	—	Structured results: <code>{glucosa: 90, cholesterol: 180, ...}</code>
6	<code>notes</code>	<code>TEXT</code>	NULL	—	Bioanalyst / lab technician notes
7	<code>reviewedBy</code>	<code>VARCHAR</code>	NULL	—	FK -> User. Doctor who reviewed results
8	<code>reviewedAt</code>	<code>TIMESTAMP</code>	NULL	—	When the doctor reviewed
9	<code>status</code>	<code>LabResultStatus</code>	NOT NULL	<code>PENDING</code>	PENDING -> COMPLETED -> ABNORMAL -> CANCELLED
10	<code>performedAt</code>	<code>TIMESTAMP</code>	NULL	—	Actual sample collection / analysis date
11	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Record creation
12	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Last update

Business rules:

- 1:1 with `LabRequest` — each lab order produces exactly one result record
- `resultJson` is flexible JSONB: structure depends on the exam type (e.g., hemogram has different fields than glicemia)
- `status=ABNORMAL` triggers a notification to the requesting doctor (values outside normal range)
- `reviewedBy` + `reviewedAt` track when the doctor acknowledged the results
- `patientId` is denormalized for dashboard queries ("show all results for patient X")

Indexes:

- PRIMARY KEY (id)
- UNIQUE (labRequestId)
- INDEX (patientId)
- INDEX (status)

Enum: LabResultStatus

VALUE	DESCRIPTION
PENDING	Awaiting lab processing
COMPLETED	Results ready, within normal range
ABNORMAL	Results contain values outside normal range — flag for doctor
CANCELLED	Lab request cancelled

Zod schema: `src/features/lab-results/schemas.ts`

13. Billing & Payments v3

Module purpose: Complete medical billing workflow. Doctors/clinics create invoices linked to consultations/prescriptions. Supports multiple line items and partial payments via different methods.

Invoice

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> User. Doctor/clinic issuing the invoice
3	<code>patientId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Patient
4	<code>consultationId</code>	<code>VARCHAR</code>	NULL	—	FK -> Consultation. Optional link
5	<code>prescriptionId</code>	<code>VARCHAR</code>	NULL	—	FK -> Prescription. Optional link
6	<code>subtotal</code>	<code>DECIMAL (10, 2)</code>	NOT NULL	<code>0</code>	Sum of line items before tax/discount
7	<code>tax</code>	<code>DECIMAL (10, 2)</code>	NULL	<code>0</code>	Tax amount
8	<code>discount</code>	<code>DECIMAL (10, 2)</code>	NULL	<code>0</code>	Discount amount
9	<code>total</code>	<code>DECIMAL (10, 2)</code>	NOT NULL	<code>0</code>	Final amount due (subtotal + tax - discount)
10	<code>currency</code>	<code>VARCHAR</code>	NOT NULL	<code>USD</code>	Currency code (ISO 4217)
11	<code>status</code>	<code>InvoiceStatus</code>	NOT NULL	<code>DRAFT</code>	DRAFT -> SENT -> PAID -> PARTIALLY_PAID -> OVERDUE -> CANCELLED
12	<code>dueDate</code>	<code>DATE</code>	NULL	—	Payment due date
13	<code>notes</code>	<code>TEXT</code>	NULL	—	Invoice notes / payment instructions
14	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW ()</code>	Creation
15	<code>updatedAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW ()</code>	Last update

Business rules:

- Status workflow: `DRAFT` (being edited) -> `SENT` (sent to patient) -> `PAID` (fully settled) / `PARTIALLY_PAID` (partial payment) / `OVERDUE` (past dueDate) / `CANCELLED` (voided)
- `total` should equal sum of `InvoiceItem.total` + `tax` - `discount`
- Multiple `Payment` records can be linked to one invoice (partial payments)
- `userId` is RESTRICT on delete — invoices cannot be orphaned

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId)`
- `INDEX (patientId)`
- `INDEX (status)`
- `INDEX (dueDate)` WHERE status IN ('SENT', 'PARTIALLY_PAID') — overdue detection

InvoiceItem

Line item within an invoice.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>invoiceId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Invoice. CASCADE delete
3	<code>description</code>	<code>VARCHAR</code>	NOT NULL	—	e.g. "Consulta general", "Amoxicilina 500mg x30"
4	<code>quantity</code>	<code>INT</code>	NOT NULL	1	Quantity of this item
5	<code>unitPrice</code>	<code>DECIMAL(10,2)</code>	NOT NULL	0	Price per unit
6	<code>total</code>	<code>DECIMAL(10,2)</code>	NOT NULL	0	quantity × unitPrice

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (invoiceId)`

Payment

Payment received against an invoice.

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>invoiceId</code>	<code>VARCHAR</code>	NOT NULL	—	FK -> Invoice. CASCADE delete

3	amount	DECIMAL(10, 2)	NOT NULL	—	Payment amount
4	method	PaymentMethod	NOT NULL	—	CASH -> CARD -> TRANSFER -> INSURANCE -> OTHER
5	reference	VARCHAR	NULL	—	Transaction reference number
6	paidAt	TIMESTAMP	NOT NULL	NOW()	Payment timestamp
7	notes	TEXT	NULL	—	Payment notes

Indexes:

- PRIMARY KEY (id)
- INDEX (invoiceId)

Enum: InvoiceStatus

VALUE	DESCRIPTION
DRAFT	Being created/edited, not yet sent
SENT	Sent to patient, awaiting payment
PAYED	Fully paid
PARTIALLY_PAID	Partial payment received
OVERDUE	Past due date without full payment
CANCELLED	Voided/cancelled invoice

Enum: PaymentMethod

VALUE	LABEL (ES)	DESCRIPTION
CASH	Efectivo	Cash payment
CARD	Tarjeta	Credit/debit card

TRAN SFER	Transferencia	Bank transfer
INSU RANC E	Seguro	Insurance payment
OTHE R	Otro	Other method

Zod schema: `src/features/billing/schemas.ts`

14. Audit Log v3

Module purpose: HIPAA/GDPR-compliant access tracking. Every view, create, update, delete, export, or print of patient data is recorded with user identity, IP address, and action details.

AuditLog

#	FIELD	TYPE	NULL	DEFAULT	DESCRIPTION
1	<code>id</code>	<code>VARCHAR</code>	NOT NULL	—	Primary key (UUID v7)
2	<code>userId</code>	<code>VARCHAR</code>	NULL	—	FK -> User. Who performed the action. NULL if user deleted
3	<code>patientId</code>	<code>VARCHAR</code>	NULL	—	FK -> Patient. Whose data was accessed. NULL for non-patient actions
4	<code>action</code>	<code>AuditAction</code>	NOT NULL	—	Type of action performed
5	<code>resource</code>	<code>VARCHAR</code>	NOT NULL	—	Resource identifier, e.g. "Consultation:abc123"
6	<code>resourceType</code>	<code>VARCHAR</code>	NOT NULL	—	Resource type, e.g. "Consultation", "Prescription"
7	<code>details</code>	<code>JSONB</code>	NULL	—	Action metadata: <code>{changedFields: [...], oldValues: {...}}</code>
8	<code>ipAddress</code>	<code>VARCHAR</code>	NULL	—	Client IP address
9	<code>userAgent</code>	<code>VARCHAR</code>	NULL	—	Client user agent string
10	<code>createdAt</code>	<code>TIMESTAMP</code>	NOT NULL	<code>NOW()</code>	Action timestamp

Business rules:

- Every VIEW of patient data MUST be logged (HIPAA access log requirement)
- CREATE/UPDATE/DELETE store changed field names and old values in `details` `JSONB` for rollback
- EXPORT logs when data is downloaded (CSV, PDF)
- PRINT logs when data is printed
- LOGIN/LOGOUT track session activity
- `userId` is SET NULL on user deletion — preserves the audit trail
- Recommended: partition table by month for performance with large volumes

Indexes:

- `PRIMARY KEY (id)`
- `INDEX (userId)`

- INDEX (patientId)
- INDEX (resourceType, resource)
- INDEX (createdAt DESC) — time-based queries

Enum: AuditAction

VALUE	DESCRIPTION
VIEW	Data was viewed/read
CREATE	New record created
UPDATE	Existing record modified
DELETE	Record deleted (soft or hard)
EXPORT	Data exported/downloaded
LOGIN	User logged in
LOGOUT	User logged out
PRINT	Data sent to printer

Zod schema: `src/features/audit/schemas.ts`

15. Entity Relationship Diagram



Relationship notation:

- ➡ : Foreign key (many-to-one)
- ➡ : One-to-one

- 1:1 : Unique constraint on FK
 - 1:N : One-to-many
-

Appendix: Quick Reference

All Enums

ENUM	MODULE	VALUES
User Role	2	DOCTOR, PROVIDER, ADMIN
Plan Type	2	FREE, PRO, ENTERPRISE
Gender	3	MALE, FEMALE, OTHER
Presentation Enum	5	CAPSULA, TABLETA, JARABE, GOTAS, AMPOLLA, CREMA
Administration Route Enum	5	ORAL, INTRAVENOSA, INTRAMUSCULAR, TOPICA, OFTALMICA
Rx Status	5	ACTIVE, CANCELLED, EXPIRED
Provider Type	6	PHARMACY, LABORATORY
Quote Status	6	OPEN, CLOSED
Notification Type	7	SYSTEM, NEW_QUOTE_REQUEST, QUOTE_RECEIVED, FOLLOW_UP_ALERT
Document Type	7	CERTIFICATE, REFERRAL, REPORT
Follow Status	7	PENDING, SENT, RESPONDED
Clinic Role	9	OWNER, ADMIN, DOCTOR, RECEPTIONIST

DocVerificationType	10	MEDICAL_LICENSE, NATIONAL_ID, BUSINESS_RIF, COMMERCIAL_REGISTER
VerificationStatus	10	PENDING, APPROVED, REJECTED
LabResultStatus	12	PENDING, COMPLETED, ABNORMAL, CANCELLED
InvoicesStatus	13	DRAFT, SENT, PAID, PARTIALLY_PAID, OVERDUE, CANCELLED
PaymentMethod	13	CASH, CARD, TRANSFER, INSURANCE, OTHER
AuditAction	14	VIEW, CREATE, UPDATE, DELETE, EXPORT, LOGIN, LOGOUT, PRINT

All SQL Patches

FILE	VERSION	CONTENTS
DATA_BASE_PAT_CH_v2.sql	v2 (2026-06-08)	Medication, Appointment, Consultation.status, PrescriptionItem updates
DATA_BASE_PAT_CH_v3.sql	v3 (2026-06-09)	PharmacyInventory, LabResult, Invoice/InvoiceItem/Payment, AuditLog, Patient/Medication improvements

Frontend Zod Schemas

FEATURE	SCHEMA FILE
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Patients	<code>src/features/patients/schemas.ts</code>
Medications	<code>src/features/medications/schemas.ts</code>
Appointments	<code>src/features/appointments/schemas.ts</code>
Consultations	<code>src/features/consultations/schemas.ts</code>
Inventory	<code>src/features/inventory/schemas.ts</code>
Lab Results	<code>src/features/lab-results/schemas.ts</code>
Billing	<code>src/features/billing/schemas.ts</code>
Audit	<code>src/features/audit/schemas.ts</code>

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